

CUSTOMER BEHAVIOR ANALYSIS REPORT

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Internship : Alfido Tech
Project Title : Customer Behavior Analysis
Tool Used : Python (Jupyter Notebook)
Dataset Source : Kaggle Customer Behavior Analysis Dataset

Executive Summary

This project analyzed customer transactions and customer behavior to identify customer segments, purchasing trends and churn risks.

The dataset was processed using Python in Jupyter Notebook. Data cleaning was performed to remove missing values and duplicates. Feature engineering techniques were applied to classify customer spending levels.

Customer segmentation helped identify different customer value groups and understand customer engagement patterns.

The analysis provides business recommendations to improve retention and customer experience.

1. Data Cleaning

The dataset was cleaned before analysis.

Cleaning activities performed:

- Missing values identified
- Null records removed
- Duplicate rows removed
- Data quality verified

Result:

Dataset became suitable for analysis and visualization.

2. Feature Engineering

Additional features were created to improve analysis.

Created Features:

- Purchase Level
 - Low
 - Medium
 - High

Purpose:

To classify customers based on spending behavior.

3. Customer Segmentation

Customers were grouped based on purchase amount.

Segments identified:

High Value Customers

Customers with high purchase activity.

Medium Value Customers

Regular customers with moderate engagement.

Low Value Customers

Customers showing low purchasing activity.

Purpose:

Understand customer behavior and improve targeting.

4. Purchase Pattern Analysis

Purchase behavior was analyzed across product categories.

Observations:

- Some categories received higher customer attention.
 - Purchase amount varied across customers.
 - Customer activity was not evenly distributed.
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5. Customer Retention and Churn Analysis

Customer churn patterns were examined.

Findings:

- Some customers showed lower retention.
 - Churn risk can affect future revenue.
 - Engagement strategies are necessary.
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Key Findings

1. Customers belong to multiple spending segments.
 2. High-value customers contribute significantly.
 3. Product preference differs across groups.
 4. Customer retention needs improvement.
 5. Targeted campaigns can increase engagement.
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5 Actionable Recommendations for Alfido Tech

Recommendation 1

Provide loyalty benefits to high-value customers.

Recommendation 2

Create personalized marketing campaigns.

Recommendation 3

Introduce customer retention programs.

Recommendation 4

Target medium-value customers with promotional offers.

Recommendation 5

Track customer churn regularly and improve engagement.

Conclusion

Customer behavior analysis provided valuable insights into customer segmentation, purchasing behavior and retention trends.

These findings can support business decisions and improve customer engagement strategies at Alfido Tech.

Deliverables Submitted

✓ Jupyter Notebook

✓ Visualizations

✓ PDF Report

✓ GitHub Repository Link